

AP[®] Physics C: Mechanics Practice Exam

From the 2014 Administration

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TABLE OF INFORMATION, EFFECTIVE 2012

CONSTANTS AND CONVERSION FACTORS	
Proton mass, $m_p = 1.67 \times 10^{-27}$ kg	Electron charge magnitude, $e = 1.60 \times 10^{-19}$ C
Neutron mass, $m_n = 1.67 \times 10^{-27}$ kg	1 electron volt, $1 \text{ eV} = 1.60 \times 10^{-19}$ J
Electron mass, $m_e = 9.11 \times 10^{-31}$ kg	Speed of light, $c = 3.00 \times 10^8$ m/s
Avogadro's number, $N_0 = 6.02 \times 10^{23}$ mol ⁻¹	Universal gravitational constant, $G = 6.67 \times 10^{-11}$ m ³ /kg·s ²
Universal gas constant, $R = 8.31$ J/(mol·K)	Acceleration due to gravity at Earth's surface, $g = 9.8$ m/s ²
Boltzmann's constant, $k_B = 1.38 \times 10^{-23}$ J/K	
1 unified atomic mass unit,	$1 \text{ u} = 1.66 \times 10^{-27}$ kg = $931 \text{ MeV}/c^2$
Planck's constant,	$h = 6.63 \times 10^{-34}$ J·s = 4.14×10^{-15} eV·s
	$hc = 1.99 \times 10^{-25}$ J·m = 1.24×10^3 eV·nm
Vacuum permittivity,	$\epsilon_0 = 8.85 \times 10^{-12}$ C ² /N·m ²
Coulomb's law constant, $k = 1/4\pi\epsilon_0 = 9.0 \times 10^9$ N·m ² /C ²	
Vacuum permeability,	$\mu_0 = 4\pi \times 10^{-7}$ (T·m)/A
Magnetic constant, $k' = \mu_0/4\pi = 1 \times 10^{-7}$ (T·m)/A	
1 atmosphere pressure,	$1 \text{ atm} = 1.0 \times 10^5$ N/m ² = 1.0×10^5 Pa

UNIT SYMBOLS	meter,	m	mole,	mol	watt,	W	farad,	F
	kilogram,	kg	hertz,	Hz	coulomb,	C	tesla,	T
	second,	s	newton,	N	volt,	V	degree Celsius,	°C
	ampere,	A	pascal,	Pa	ohm,	Ω	electron-volt,	eV
	kelvin,	K	joule,	J	henry,	H		

PREFIXES		
Factor	Prefix	Symbol
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

VALUES OF TRIGONOMETRIC FUNCTIONS FOR COMMON ANGLES							
θ	0°	30°	37°	45°	53°	60°	90°
$\sin \theta$	0	1/2	3/5	$\sqrt{2}/2$	4/5	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	4/5	$\sqrt{2}/2$	3/5	1/2	0
$\tan \theta$	0	$\sqrt{3}/3$	3/4	1	4/3	$\sqrt{3}$	∞

The following conventions are used in this exam.

- Unless otherwise stated, the frame of reference of any problem is assumed to be inertial.
- The direction of any electric current is the direction of flow of positive charge (conventional current).
- For any isolated electric charge, the electric potential is defined as zero at an infinite distance from the charge.

ADVANCED PLACEMENT PHYSICS C EQUATIONS, EFFECTIVE 2012

MECHANICS		ELECTRICITY AND MAGNETISM	
$v = v_0 + at$	$a = \text{acceleration}$	$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$	$A = \text{area}$
$x = x_0 + v_0 t + \frac{1}{2} at^2$	$F = \text{force}$	$\mathbf{E} = \frac{\mathbf{F}}{q}$	$B = \text{magnetic field}$
$v^2 = v_0^2 + 2a(x - x_0)$	$f = \text{frequency}$	$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q}{\epsilon_0}$	$C = \text{capacitance}$
$\Sigma \mathbf{F} = \mathbf{F}_{net} = m\mathbf{a}$	$h = \text{height}$	$E = -\frac{dV}{dr}$	$d = \text{distance}$
$\mathbf{F} = \frac{d\mathbf{p}}{dt}$	$I = \text{rotational inertia}$	$V = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i}$	$E = \text{electric field}$
$\mathbf{J} = \int \mathbf{F} dt = \Delta \mathbf{p}$	$J = \text{impulse}$	$U_E = qV = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$	$\mathcal{E} = \text{emf}$
$\mathbf{p} = m\mathbf{v}$	$K = \text{kinetic energy}$	$C = \frac{Q}{V}$	$F = \text{force}$
$F_{fric} \leq \mu N$	$k = \text{spring constant}$	$C = \frac{\kappa \epsilon_0 A}{d}$	$I = \text{current}$
$W = \int \mathbf{F} \cdot d\mathbf{r}$	$\ell = \text{length}$	$C_p = \sum_i C_i$	$J = \text{current density}$
$K = \frac{1}{2} mv^2$	$L = \text{angular momentum}$	$\frac{1}{C_s} = \sum_i \frac{1}{C_i}$	$L = \text{inductance}$
$P = \frac{dW}{dt}$	$m = \text{mass}$	$I = \frac{dQ}{dt}$	$\ell = \text{length}$
$P = \mathbf{F} \cdot \mathbf{v}$	$N = \text{normal force}$	$U_c = \frac{1}{2} QV = \frac{1}{2} CV^2$	$n = \text{number of loops of wire per unit length}$
$\Delta U_g = mgh$	$P = \text{power}$	$\oint \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I$	$N = \text{number of charge carriers per unit volume}$
$a_c = \frac{v^2}{r} = \omega^2 r$	$p = \text{momentum}$	$R = \frac{\rho \ell}{A}$	$P = \text{power}$
$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$	$r = \text{radius or distance}$	$\mathbf{E} = \rho \mathbf{J}$	$Q = \text{charge}$
$\Sigma \boldsymbol{\tau} = \boldsymbol{\tau}_{net} = I\boldsymbol{\alpha}$	$\mathbf{r} = \text{position vector}$	$I = Nev_d A$	$q = \text{point charge}$
$I = \int r^2 dm = \Sigma mr^2$	$T = \text{period}$	$V = IR$	$R = \text{resistance}$
$\mathbf{r}_{cm} = \Sigma m\mathbf{r} / \Sigma m$	$U = \text{potential energy}$	$R_s = \sum_i R_i$	$r = \text{distance}$
$v = r\omega$	$v = \text{velocity or speed}$	$\frac{1}{R_p} = \sum_i \frac{1}{R_i}$	$t = \text{time}$
$\mathbf{L} = \mathbf{r} \times \mathbf{p} = I\boldsymbol{\omega}$	$W = \text{work done on a system}$	$P = IV$	$U = \text{potential or stored energy}$
$K = \frac{1}{2} I\omega^2$	$x = \text{position}$	$\mathbf{F}_M = q\mathbf{v} \times \mathbf{B}$	$V = \text{electric potential}$
$\omega = \omega_0 + \alpha t$	$\mu = \text{coefficient of friction}$		$v = \text{velocity or speed}$
$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$	$\theta = \text{angle}$		$\rho = \text{resistivity}$
	$\tau = \text{torque}$		$\phi_m = \text{magnetic flux}$
	$\omega = \text{angular speed}$		$\kappa = \text{dielectric constant}$
	$\alpha = \text{angular acceleration}$		
	$\phi = \text{phase angle}$		
	$\mathbf{F}_s = -k\mathbf{x}$		
	$U_s = \frac{1}{2} kx^2$		
	$x = x_{\max} \cos(\omega t + \phi)$		
	$T = \frac{2\pi}{\omega} = \frac{1}{f}$		
	$T_s = 2\pi \sqrt{\frac{m}{k}}$		
	$T_p = 2\pi \sqrt{\frac{\ell}{g}}$		
	$\mathbf{F}_G = -\frac{Gm_1 m_2}{r^2} \hat{\mathbf{r}}$		
	$U_G = -\frac{Gm_1 m_2}{r}$		

ADVANCED PLACEMENT PHYSICS C EQUATIONS, EFFECTIVE 2012

GEOMETRY AND TRIGONOMETRY

Rectangle

$$A = bh$$

Triangle

$$A = \frac{1}{2}bh$$

Circle

$$A = \pi r^2$$

$$C = 2\pi r$$

Rectangular Solid

$$V = \ell wh$$

Cylinder

$$V = \pi r^2 \ell$$

$$S = 2\pi r \ell + 2\pi r^2$$

Sphere

$$V = \frac{4}{3}\pi r^3$$

$$S = 4\pi r^2$$

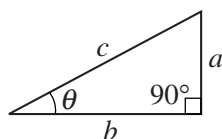
Right Triangle

$$a^2 + b^2 = c^2$$

$$\sin \theta = \frac{a}{c}$$

$$\cos \theta = \frac{b}{c}$$

$$\tan \theta = \frac{a}{b}$$



CALCULUS

$$\frac{df}{dx} = \frac{df}{du} \frac{du}{dx}$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1$$

$$\int e^x dx = e^x$$

$$\int \frac{dx}{x} = \ln|x|$$

$$\int \cos x dx = \sin x$$

$$\int \sin x dx = -\cos x$$

PHYSICS C: MECHANICS

SECTION I

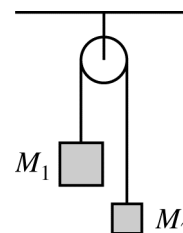
Time—45 minutes

35 Questions

Directions: Each of the questions or incomplete statements below is followed by five suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.

Note: To simplify calculations, you may use $g = 10\text{ m/s}^2$ in all problems.

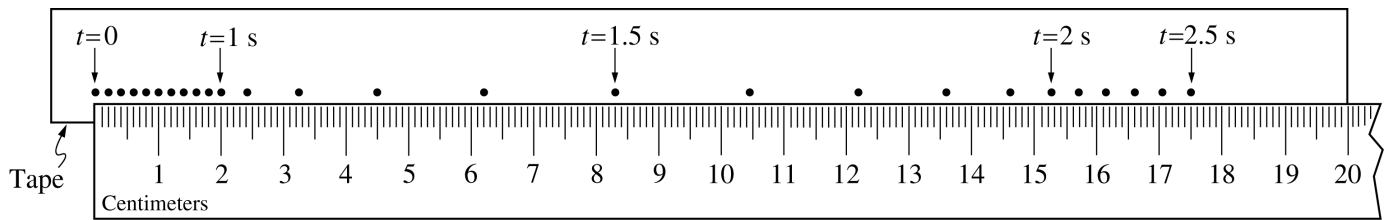
1. An object is released from rest from a great height and reaches its terminal velocity. Which of the following statements is true of the object while it is falling with terminal velocity?
 - (A) There is no longer a gravitational force on it.
 - (B) There is no longer a drag (air resistance) force on it.
 - (C) Its acceleration is upward.
 - (D) The magnitudes of the gravitational and drag forces on it are equal.
 - (E) The gravitational and drag forces on it act in the same direction.
2. A student with a mass of 50 kg is standing on a bathroom scale while riding in an elevator. If the reading on the scale is 400 N, which of the following is a correct description of the elevator's motion?
 - (A) Moving upward with increasing speed
 - (B) Moving upward with constant speed
 - (C) Moving downward with constant speed
 - (D) Moving downward with increasing speed
 - (E) Moving downward with decreasing speed



3. The system represented above consists of two objects of unequal masses, M_1 and M_2 , with $M_1 > M_2$. The objects hang from the ends of a cord of negligible mass that passes over a pulley with negligible mass and friction. Which of the following is true about the changes in the gravitational potential energy, ΔU , and kinetic energy, ΔK , of the system soon after the objects are released from rest?
 - (A) $\Delta U < 0$ and $\Delta K > 0$
 - (B) $\Delta U = 0$ and $\Delta K > 0$
 - (C) $\Delta U < 0$ and $\Delta K = 0$
 - (D) $\Delta U = 0$ and $\Delta K = 0$
 - (E) $\Delta U > 0$ and $\Delta K < 0$

Questions 4-6

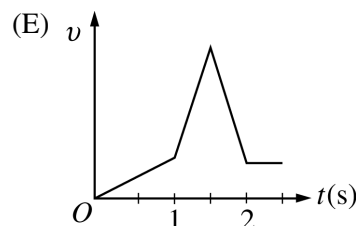
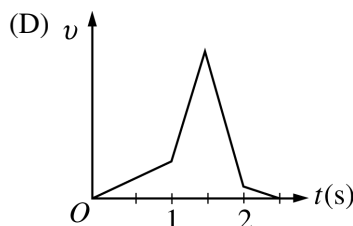
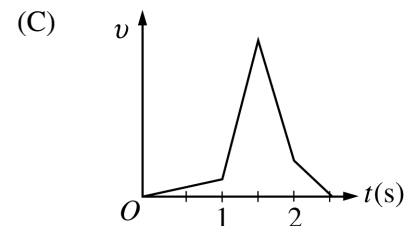
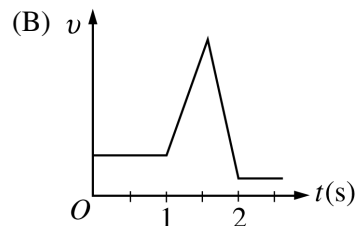
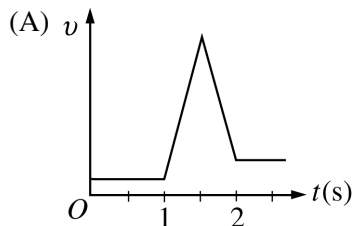
A tape attached to a moving object was pulled by the object through a marker that put dots on the tape at a constant rate of 10 dots per second for a period of 2.5 s. The figure below shows the marked tape next to a centimeter ruler.



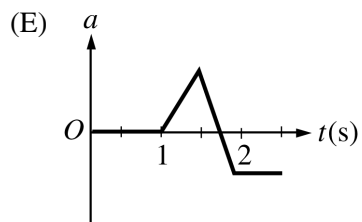
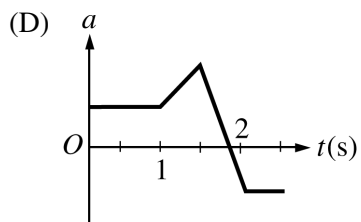
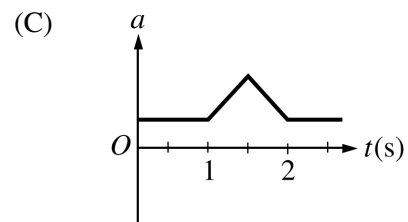
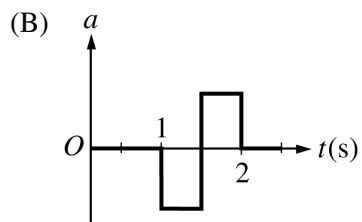
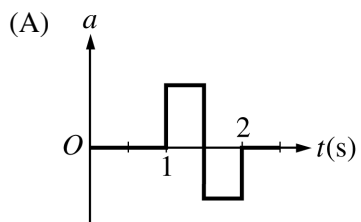
4. The average speed of the object for the total time recorded on the tape is most nearly

- (A) 2.0 cm/s
- (B) 3.3 cm/s
- (C) 4.5 cm/s
- (D) 5.5 cm/s
- (E) 7.0 cm/s

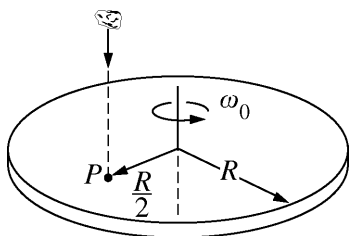
5. Which of the following best represents the graph of the velocity of the object versus time?



6. Which of the following best represents the graph of the acceleration of the object versus time?



Questions 7-8



A turntable with mass m , radius R , and rotational inertia $\frac{mR^2}{2}$ initially rotates freely about an axis through its center at constant angular speed with negligible friction. A piece of clay, also of mass m , falls vertically onto the turntable, as shown above, and sticks to it at point P , a distance $\frac{R}{2}$ from the center of rotation.

7. What is the rotational inertia of the clay-turntable system after the collision?

(A) $\frac{1}{4}mR^2$
 (B) $\frac{1}{2}mR^2$
 (C) $\frac{3}{4}mR^2$
 (D) mR^2
 (E) $2mR^2$

8. What happens to the rotational speed of the turntable and the angular momentum of the clay-turntable system about the axis as a result of the collision?

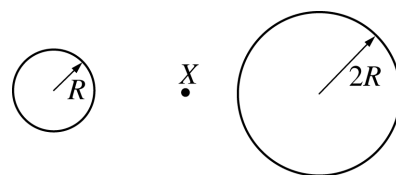
<u>Rotational Speed</u>	<u>Angular Momentum</u>
(A) Stays the same	Increases
(B) Stays the same	Stays the same
(C) Stays the same	Decreases
(D) Decreases	Stays the same
(E) Decreases	Decreases

9. Which of the following must be true in order for a rotating platform to continue rotating with a constant angular velocity?

(A) There are no forces exerted on it.
 (B) There is no friction exerted on it.
 (C) There is zero net force exerted on it.
 (D) There are no torques exerted on it.
 (E) There is zero net torque exerted on it.

10. A newly discovered planet is found to have twice the radius and three times the mass of Earth. If the acceleration due to gravity at the surface of Earth is g , the acceleration due to gravity at the surface of the new planet is

(A) $\frac{2g}{3}$
 (B) $\frac{3g}{4}$
 (C) g
 (D) $\frac{4g}{3}$
 (E) $\frac{3g}{2}$

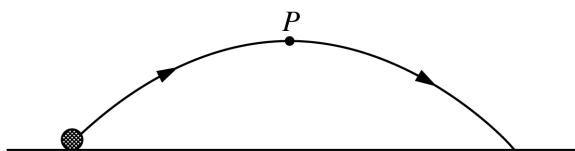


11. At point X shown above, which is midway between the centers of two isolated planets of radii R and $2R$, the net gravitational force on an object is zero. If the mass of the smaller planet is M , the mass of the larger planet is

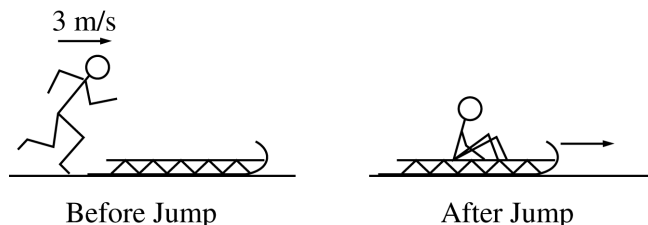
(A) $\frac{M}{4}$
 (B) $\frac{M}{2}$
 (C) M
 (D) $2M$
 (E) $4M$

12. A particle moves along a straight line. Its speed in m/s is given by $v = a + bt^2$, where a and b are constants and t is time in seconds. How far does the particle move between $t = 1$ s and $t = 2$ s?

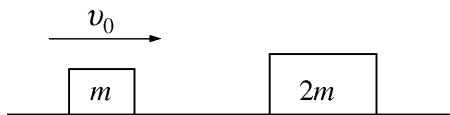
- (A) $2b$
 (B) $a + b$
 (C) $a + 2b$
 (D) $a + \frac{7}{3}b$
 (E) $a + \frac{7}{2}b$



13. A ball is tossed and follows the trajectory shown above. Point P is the highest point on the trajectory, and air resistance is negligible. Which of the following statements is true of the ball while it is in motion?
- (A) Its speed is constant.
 (B) Its acceleration is constant.
 (C) The horizontal component of its velocity is zero at point P .
 (D) The vertical component of its velocity is a maximum at point P .
 (E) Its kinetic energy is a maximum at point P .

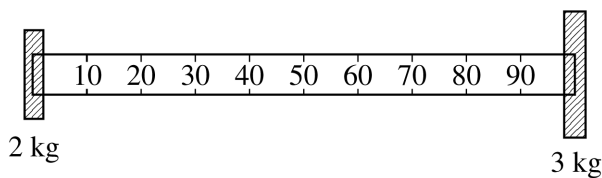


14. As shown in the figure above, a child of mass 25 kg who is running at a speed of 3 m/s jumps onto a stationary sled of mass 5 kg on a frozen lake. The speed at which the child and sled begin to slide across the ice is most nearly
- (A) 0.4 m/s
 (B) 0.6 m/s
 (C) 1.3 m/s
 (D) 2.5 m/s
 (E) 15 m/s

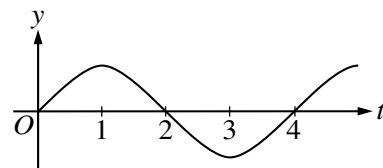


15. A block of mass m is initially sliding with speed v_0 on a horizontal frictionless surface, as shown above. It makes an elastic, head-on collision with another block of mass $2m$ that is initially at rest. Which of the following correctly shows the motion of the blocks after the collision?

- (A)
- (B)
- (C)
- (D)
- (E)

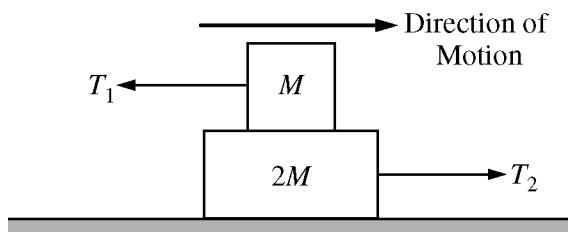


16. A 2 kg mass is attached to the 0 cm mark of a lightweight meter stick, and a 3 kg mass is attached to the 100 cm mark, as shown above. The center of mass of the combined system (meter stick and both masses) is located nearest which of the following marks on the meter stick?
- (A) 33 cm
(B) 40 cm
(C) 60 cm
(D) 67 cm
(E) 75 cm



17. For a mass attached to a vertical spring, the vertical displacement y of the mass is given as a function of time t in the diagram above. At what time t does the mass experience a maximum positive acceleration?
- (A) $t = 1$
(B) $t = 2$
(C) $t = 3$
(D) $t = 4$
(E) None of the above, since its acceleration is constant
18. For an object in simple harmonic motion with amplitude A , the kinetic energy will equal the potential energy when the displacement is
- (A) zero
(B) $\pm A/4$
(C) $\pm A/2$
(D) $\pm A/\sqrt{2}$
(E) $\pm A$
19. Examples of simple harmonic motion include which of the following?
- I. A mass, hanging from a spring, moves up and down in a uniform gravitational field.
 - II. A puck on an air table bounces back and forth between the walls of the table.
 - III. A ball bounces up and down on a table.
- (A) I only
(B) III only
(C) I and II only
(D) II and III only
(E) I, II, and III

Questions 20-21



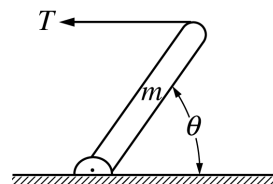
Two blocks are stacked on top of each other, as shown in the figure above. The top block, of mass M , is attached to a rope that pulls to the left with tension T_1 . The bottom block, of mass $2M$, is attached to a rope that pulls to the right with tension T_2 . The coefficient of friction between the blocks is μ . The blocks are at rest relative to each other and move to the right at constant speed relative to the frictionless surface.

20. What is the relationship between T_1 and T_2 ?

- (A) $T_1 = 2T_2$
- (B) $T_1 = \frac{1}{2}T_2$
- (C) $T_1 = \mu T_2$
- (D) $T_1 = \frac{1}{\mu}T_2$
- (E) $T_1 = T_2$

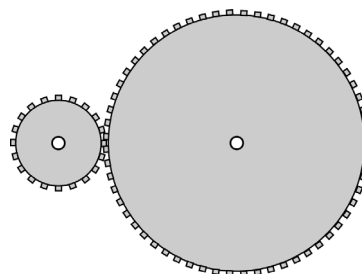
21. The force of friction experienced by the top block is f_1 , and the force of friction experienced by the bottom block is f_2 . What is the relationship between f_1 and f_2 ?

- (A) $f_1 = 2f_2$
- (B) $f_1 = \frac{1}{2}f_2$
- (C) $f_1 = \mu f_2$
- (D) $f_1 = \frac{1}{\mu}f_2$
- (E) $f_1 = f_2$



22. A uniform beam of mass m is pivoted at one end and held in equilibrium at an angle θ by a horizontal wire, as shown above. The tension T in the wire is

- (A) $\frac{1}{2}mg \cot \theta$
- (B) $mg \cot \theta$
- (C) $mg \tan \theta$
- (D) $2mg \tan \theta$
- (E) mg



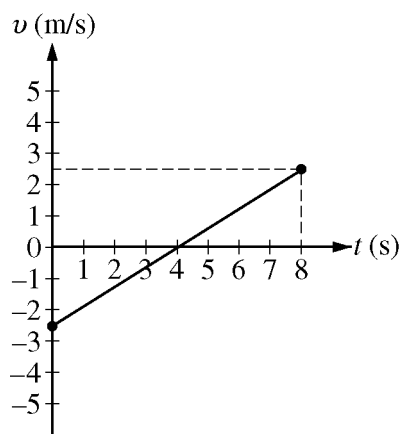
23. A gear containing 20 teeth on its circumference is meshed with a gear containing 60 teeth on its circumference, as shown above. If the angular speed of the 20-tooth gear is ω , what is the angular speed of the 60-tooth gear?

- (A) $\frac{\omega}{60}$
- (B) $\frac{\omega}{20}$
- (C) $\frac{\omega}{3}$
- (D) ω
- (E) 3ω

24. The angular displacement θ of a rotating wheel is described by the equation $\theta = \theta_0 + at^2 - bt^3$, where t is time and θ_0 , a , and b are positive constants. The angular acceleration of the wheel as a function of time t is

- (A) equal to a positive constant
- (B) zero
- (C) $2a - 6bt$
- (D) $2at - 3bt^2$
- (E) $\theta_0 t + \frac{1}{3}at^3 - \frac{1}{4}bt^4$

Questions 25-27



The motion of an object moving along a straight line is described by the graph of velocity v versus time t above.

25. Which of the following is true about the speed of the object during the time interval from 0 to 8 s?
- (A) It increases during the entire interval.
 - (B) It decreases during the entire interval.
 - (C) It remains the same.
 - (D) It first increases and then decreases.
 - (E) It first decreases and then increases.

26. Which of the following is true about the acceleration of the object during the time interval from 0 to 8 s?

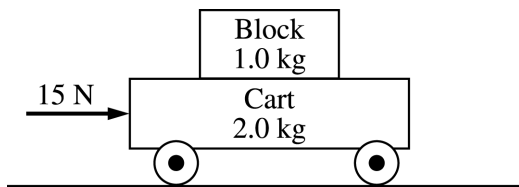
- (A) It increases during the entire interval.
- (B) It decreases during the entire interval.
- (C) It remains the same.
- (D) It first increases and then decreases.
- (E) It first decreases and then increases.

27. What is the net displacement of the object for the time interval from 0 to 8 s?

- (A) 0 m
- (B) 5 m
- (C) 10 m
- (D) 20 m
- (E) It cannot be determined without knowing the initial position of the object.

-
28. An object starts from rest and accelerates uniformly in the horizontal direction, moving 32 m during the first 4 s. What are the magnitudes of the object's displacement and velocity after it accelerates for the first 3 s

- | | Magnitude of
Displacement | Magnitude of
Velocity |
|-----|------------------------------|--------------------------|
| (A) | 24 m | 24 m/s |
| (B) | 24 m | 18 m/s |
| (C) | 18 m | 18 m/s |
| (D) | 18 m | 12 m/s |
| (E) | 8 m | 12 m/s |



29. A 2.0 kg cart rolls with negligible friction on a horizontal surface, as shown in the figure above. The coefficient of static friction between the cart and the 1.0 kg block on top of the cart is 0.7. If a horizontal 15 N force is applied to the cart as shown and the block does not slide on the cart, the magnitude of the horizontal component of the force that the block exerts on the cart is most nearly

(A) 2 N
(B) 5 N
(C) 8 N
(D) 10 N
(E) 15 N

30. A car rounds a flat curve of radius R with a speed of v_0 . The coefficient of friction between the tires and the road is μ . For the car not to slide on the road, which of the following statements must be true?

(A) The car's speed v_0 cannot be greater than $\mu g R$.
(B) The car's speed v_0 cannot be less than $\mu g R$.
(C) The radius R of the car's path cannot be greater than $v_0^2 / \mu g$.
(D) The radius R of the car's path cannot be less than $v_0^2 / \mu g$.
(E) The centripetal acceleration must be less than g .

31. If two balls collide and there are no external forces acting on the balls, quantities that must be the same for the two-ball system before and after the collision include which of the following?

I. Total kinetic energy
II. Total momentum
III. Momentum in the x -direction

(A) I only
(B) III only
(C) I and II only
(D) II and III only
(E) I, II, and III

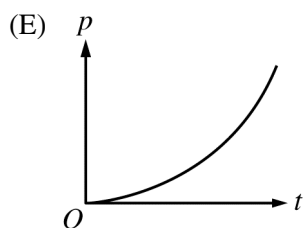
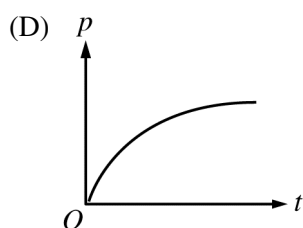
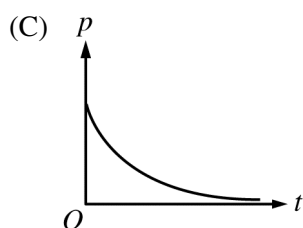
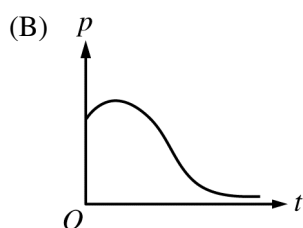
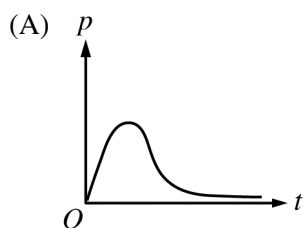
32. Two objects, X and Y , are released from rest at the same height. Object X has three times the mass of object Y . Just before reaching the ground, object Y has momentum p_Y and kinetic energy K_Y . If air resistance is negligible, what are the momentum and kinetic energy of object X just before it reaches the ground?

	<u>Momentum</u>	<u>Kinetic Energy</u>
(A)	p_Y	K_Y
(B)	$3p_Y$	$3K_Y$
(C)	$3p_Y$	$6K_Y$
(D)	$6p_Y$	$3K_Y$
(E)	$6p_Y$	$6K_Y$

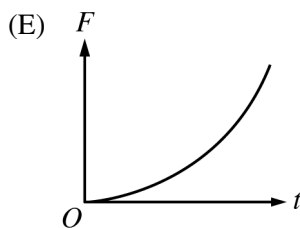
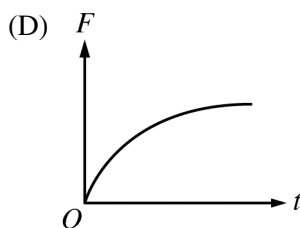
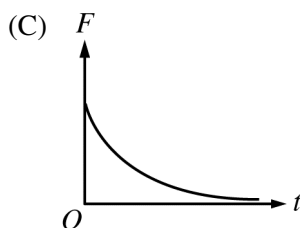
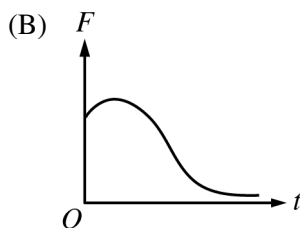
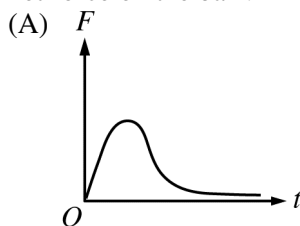
Questions 33-34

A very light ball is released from rest at time $t = 0$ and falls through the air, which provides significant resistance.

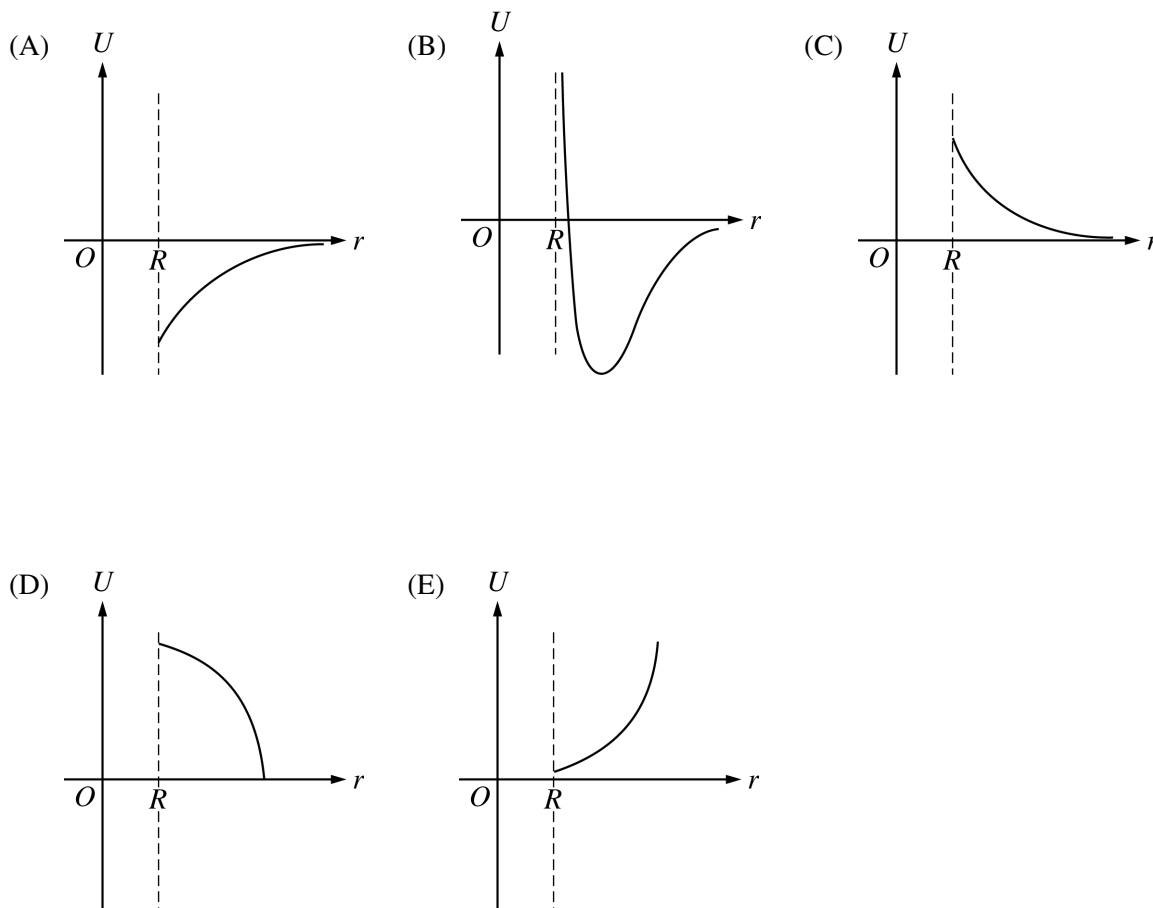
33. Which graph best represents the magnitude of the momentum of the ball?



34. Which graph best represents the magnitude of the net force on the ball?



35. A satellite can move in a circular orbit about a massive planet of radius R . Which of the following graphs could represent the gravitational potential energy U of this satellite as a function of the radius r of the orbit?



STOP

END OF MECHANICS SECTION I

**IF YOU FINISH BEFORE TIME IS CALLED,
YOU MAY CHECK YOUR WORK ON MECHANICS SECTION I ONLY.**

DO NOT TURN TO ANY OTHER TEST MATERIALS.

MAKE SURE YOU HAVE DONE THE FOLLOWING.

- **PLACED YOUR AP NUMBER LABEL ON YOUR ANSWER SHEET**
- **WRITTEN AND GRIDDED YOUR AP NUMBER CORRECTLY ON YOUR ANSWER SHEET**
- **TAKEN THE AP EXAM LABEL FROM THE FRONT OF THIS BOOKLET AND PLACED IT ON YOUR ANSWER SHEET**

PHYSICS C: MECHANICS

SECTION II

Time—45 minutes

3 Questions

Directions: Answer all three questions. The suggested time is about 15 minutes for answering each of the questions, which are worth 15 points each. The parts within a question may not have equal weight. Show all your work in this booklet in the spaces provided after each part.

Mech 1.

Experiment 1: A block of mass 1.5 kg is placed on a long board. You are to design an experiment to determine the coefficient of static friction between the block and the board.

(a)

- i. From the following list of available equipment, check those additional items you would use for the purpose of determining the coefficient of static friction.

____ Ruler

____ Spring scale

____ String

____ Meterstick

____ Pulley

____ Protractor

____ Photogate

____ Stopwatch

____ Mass hanger

____ Clamps and supports

____ Objects of various known masses

- ii. Sketch a diagram of your experimental setup and label the pieces of equipment that would be used.

- iii. Outline the experimental procedure you would use, including a list of quantities you would measure. For each quantity, identify the equipment you would use to make the measurement.

- (b) Explain how to use the measurements described in part (a) to calculate the coefficient of static friction. Include a free-body diagram in your explanation that shows all forces (not components) acting on the block while the measurements are being made.

Experiment 2: In a second experiment, the coefficient of kinetic friction between the block and the board is determined to be 0.10. The board is now inclined at an angle of 25° above the horizontal. The block is released from rest at the top of the incline and slides 2.0 m down the incline.

- (c) Calculate the work done by kinetic friction as the block slides down the incline.

- (d) The mass of the block is now increased without changing the coefficient of kinetic friction, and experiment 2 is repeated. How does each of the following change?

- i. The magnitude of the frictional force

☐ Increases ☐ Decreases ☐ Remains the same

- ii. The magnitude of the velocity of the block as it reaches the bottom of the incline

☐ Increases ☐ Decreases ☐ Remains the same

- iii. The kinetic energy of the block at the bottom of the incline

☐ Increases ☐ Decreases ☐ Remains the same

Mech 2.

A satellite of mass m is in a stable circular orbit around Earth at a distance R_1 from the center of Earth.

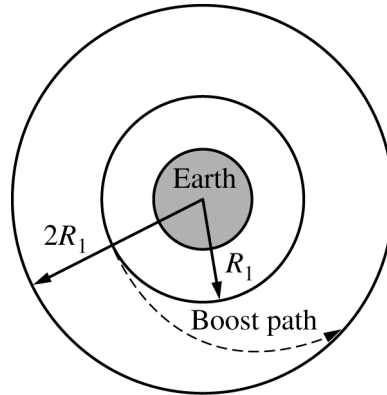
The mass of Earth is M_e .

(a) Derive an expression for the following in terms of m , R_1 , M_e , and fundamental constants, as appropriate.

i. The orbital speed v_1 of the satellite

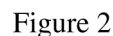
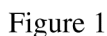
ii. The total energy of the satellite in this orbit, assuming gravitational potential energy to be zero at an infinite distance from the center of Earth

The satellite's booster rockets fire and lift the satellite to a higher circular orbit of radius $2R_1$. The satellite follows the path shown in the diagram below, moving a total distance S during the orbital change. The component of the rockets' force parallel to the path is given by the equation $F = F_0 \left(1 - \frac{x}{S}\right)$, where x is the variable distance traveled along the path at any moment.



(b) Derive an expression for the total work done on the satellite by the force F in terms of F_0 and S .

(c) If the total distance S is equal to $3R_1$, derive an expression for F_0 in terms of M_e , R_1 , m , and fundamental constants, as appropriate.



A uniform disk of mass M and radius R is suspended vertically from a pivot on its edge so that it is free to swing without friction, as shown in Figure 1 above. The disk is raised to an initial angular displacement θ_0 with respect to the vertical, as shown in Figure 2 above, and released from rest. The rotational inertia of the disk about its center of mass is $I = MR^2/2$. Express all algebraic answers in terms of the given quantities and fundamental constants, as appropriate.

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any part of this page is illegal.

(d) Derive an expression for the period for small-amplitude oscillations of the disk.

(e) Suppose the disk is replaced with another uniform disk with twice the radius, and the pivot point is again on the edge of the disk. How will the period of small-amplitude oscillations be affected by the new disk?

☐ It will increase. ☐ It will decrease. ☐ It will remain the same.

☐ There is not enough information to determine the period.

Justify your answer.

**Answer Key for AP Physics C: Mechanics
Practice Exam, Section I**

Question 1: D	Question 19: A
Question 2: D	Question 20: E
Question 3: A	Question 21: E
Question 4: E	Question 22: A
Question 5: A	Question 23: C
Question 6: A	Question 24: C
Question 7: C	Question 25: E
Question 8: D	Question 26: C
Question 9: E	Question 27: A
Question 10: B	Question 28: D
Question 11: C	Question 29: B
Question 12: D	Question 30: D
Question 13: B	Question 31: D
Question 14: D	Question 32: B
Question 15: B	Question 33: D
Question 16: C	Question 34: C
Question 17: C	Question 35: A
Question 18: D	

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Question 1

15 points total

**Distribution
of points**

(a)

i. 1 point

For selecting equipment consistent with the diagram drawn in part (a)-ii

1 point

ii. 1 point

For labeling a diagram consistent with the procedure in described part (a)-iii

1 point

iii. 2 points

For listing specific quantities to be measured that can be used to determine the coefficient of static friction

1 point

For indicating which equipment will be used to take the measurements listed above

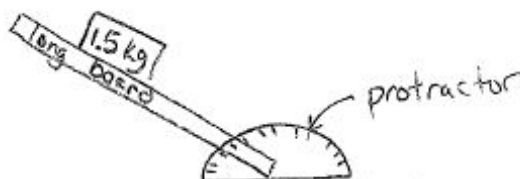
1 point

Example:



Pull the spring scale to the right until the block starts to move. Use the spring scale to measure the maximum force exerted on the block.

Alternate Example:



Place the block on the board while the board is horizontal. Lift one end of the board until the block begins to slide. Use a protractor to measure the angle at which the block begins to slide.

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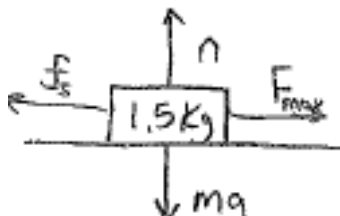
Question 1 (continued)

**Distribution
of points**

(b) 5 points

The experiment described by the student must attempt to measure the coefficient of static friction μ_s , not the coefficient of kinetic friction.

Solution for example shown above:



For vectors indicating the normal force up and the force of gravity down

1 point

For a vector indicating the horizontal applied force

1 point

For a vector indicating the static friction opposite the applied force

1 point

Note: One point is deducted if there are any extraneous vectors or any vectors not starting on and pointing away from the block.

For setting the net force equal to zero

1 point

$$F_{net} = ma = 0$$

$$F_{max} - f_s = 0$$

$$f_s = \mu_s mg$$

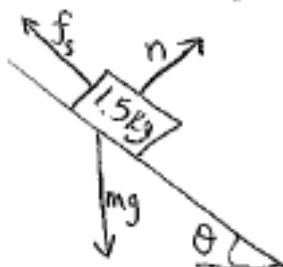
For a correct expression for the coefficient of static friction

1 point

$$\mu_s = \frac{F_{max}}{mg}$$

Solution for alternate example:

Alternate points



For a vector indicating the force of gravity down

1 point

For a vector indicating the normal force perpendicular to the incline

1 point

For a vector indicating the static friction up the incline

1 point

Note: One point is deducted if there are any extraneous vectors or any vectors not starting on and pointing away from the block

For setting the net force equal to zero

1 point

$$F_{net} = ma = 0$$

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Question 1 (continued)

**Distribution
of points**

$$F_{\max} - f_s = 0$$

$F_{\max}$ is the component of gravity along the plane, and f_s depends on the component of gravity perpendicular to the plane

$$mg \sin \theta - \mu_s mg \cos \theta = 0$$

For a correct expression for the coefficient of static friction

$$\mu_s = \tan \theta$$

1 point

(c) 3 points

For a correct equation to solve for the work done

1 point

$$W = F \cdot d = -f_k d$$

$$W = -\mu_k mg \cos \theta d$$

For correctly substituting for the force of kinetic friction

1 point

$$W = -(0.10)(1.5 \text{ kg})(9.8 \text{ m/s}^2)(\cos 25)(2.0 \text{ m})$$

For a correct answer

1 point

$$W = -2.7 \text{ J}$$

Note: Answers without the negative sign only lose the answer point.

(d)

i. 1 point

For selecting “Increases”

1 point

ii. 1 point

For selecting “Remains the same”

1 point

iii. 1 point

For selecting “Increases”

1 point

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Question 2

15 points total

**Distribution
of points**

(a)

i. 3 points

Set the gravitational force equal to the centripetal force.

$$F_g = F_C$$

For a correct expression for the gravitational force

1 point

For a correct expression for the centripetal force

1 point

$$\frac{GM_em}{R_1^2} = \frac{mv_1^2}{R_1}$$

For a correct answer

1 point

$$v_1 = \sqrt{\frac{GM_e}{R_1}}$$

ii. 3 points

For an expression of the total energy as the sum of the gravitational potential energy and the kinetic energy of the satellite

1 point

$$E = U_g + K$$

For substituting a correct expression for the kinetic energy of the satellite

1 point

$$E = -\frac{GM_em}{R_1} + \frac{1}{2}mv_1^2$$

$$E = -\frac{GM_em}{R_1} + \frac{1}{2}m\left(\frac{GM_e}{R_1}\right)$$

For a correct answer

1 point

$$E = -\frac{GM_em}{2R_1}$$

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Question 2 (continued)

**Distribution
of points**

(b) 4 points

For a correct equation for the work done expressed as the integral of the force

1 point

$$W = \int F \cdot ds$$

For substituting the given expression for force into the above equation

1 point

$$W = \int_0^S F_0 \left(1 - \frac{x}{S} \right) dx$$

For integrating with the correct limits

1 point

$$W = F_0 \left[x - \frac{x^2}{2S} \right]_0^S$$

$$W = F_0 \left(S - \frac{S^2}{2S} - 0 \right) = F_0 \left(\frac{S}{2} \right)$$

For a correct answer

1 point

$$W = \frac{F_0 S}{2}$$

(c) 5 points

For a correct statement relating work and change in energy

1 point

$$W = \Delta E$$

For substituting the correct initial radius into the expression for potential energy

1 point

For substituting the correct final radius into the expression for potential energy

1 point

$$W = \Delta E = -\frac{GM_e m}{4R_1} - \left(-\frac{GM_e m}{2R_1} \right)$$

For a correct substitution of the answer from part (b)

1 point

$$\frac{F_0 S}{2} = -\frac{GM_e m}{4R_1} + \frac{GM_e m}{2R_1}$$

$$\frac{F_0 (3R_1)}{2} = \frac{GM_e m}{4R_1}$$

For an answer consistent with part (b) and the change in energy

1 point

$$F_0 = \frac{GM_e m}{6R_1^2}$$

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Question 3

15 points total

**Distribution
of points**

(a) 2 points

For a correct expression of the parallel axis theorem

$$I = I_{CM} + Md^2$$

Substitute into the above expression

$$I = \frac{1}{2}MR^2 + MR^2$$

For a correct answer

$$I = \frac{3}{2}MR^2$$

1 point

1 point

(b) 4 points

For a correct expression of the conservation of energy for the disk

$$K_1 + U_1 = K_2 + U_2$$

For correctly setting the initial gravitational potential energy equal to the rotational kinetic energy at the bottom of the swing

$$mgh_1 = \frac{1}{2}I\omega^2$$

For substituting a correct expression for the initial height

$$Mg(R - R\cos\theta_0) = \frac{1}{2}\left(\frac{3}{2}MR^2\right)\omega^2$$

$$g(1 - \cos\theta_0) = \frac{3}{4}R\omega^2$$

For an answer consistent with part (a)

$$\omega = \sqrt{\frac{4g}{3R}(1 - \cos\theta_0)}$$

1 point

1 point

1 point

1 point

(c) 3 points

For using a correct expression of Newton's 2nd law for rotational motion

$$\tau = I\alpha = \frac{3}{2}MR^2\alpha$$

For substituting a correct expression for torque into the above equation

$$-Mg\sin\theta R = \frac{3}{2}MR^2\alpha$$

For expressing the above equation as a differential equation (Note: Any appropriate manipulation of this equation is acceptable, including a subsequent small angle approximation.)

$$-g\sin\theta = \frac{3}{2}R\frac{d^2\theta}{dt^2}$$

1 point

1 point

1 point

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Question 3 (continued)

**Distribution
of points**

(d) 4 points

For using a correct expression for simple harmonic motion

1 point

$$a = -\omega^2 x$$

$$\frac{d^2\theta}{dt^2} = -\omega^2\theta$$

For substituting the small angle approximation ($\sin\theta \approx \theta$) into the answer from part (c)

1 point

$$-g\theta = \frac{3}{2}R\frac{d^2\theta}{dt^2}$$

$$-\frac{2g}{3R}\theta = \frac{d^2\theta}{dt^2}$$

$$\omega = \sqrt{\frac{2g}{3R}}$$

For relating the correct expression for the angular velocity to the equation for period

1 point

$$\omega = \sqrt{\frac{2g}{3R}} = \frac{2\pi}{T}$$

For a correct answer

1 point

$$T = 2\pi\sqrt{\frac{3R}{2g}}$$

(e) 2 points

For selecting “It will increase”

1 point

For a correct justification

1 point

Example:

The period of the disk is proportional to the square root of the radius of the disk,

$T \propto \sqrt{R}$. A doubling of the radius of the disk will increase the period by a factor of $\sqrt{2}$.